

functionality in many applications, offering an alternative to more traditional systems like GPS, which can falter in indoor environments.

A Wi-Fi locationing system integrates several key components to achieve accurate geolocation. First, the Wi-Fi radio in the user's device detects the signal of nearby networks. Next, a comprehensive database stores mappings of Wi-Fi network identifiers, like MAC addresses, to specific geographic locations. This database is maintained and updated by location service providers to ensure accuracy and reliability. Finally, the system employs sophisticated algorithms that analyse the patterns in the database, calculate signal strengths, and correlate this information to estimate the device's location.

When a device seeks to determine its location, it scans for available Wi-Fi networks and collects data on them. This data is sent to the location service, which then checks the information against its database. By assessing the similarities in network data and analysing signal strengths, the service can effectively estimate the device's location. This technology not only enhances user experience by providing location-based services but also plays a crucial role in various practical applications, such as indoor navigation, asset tracking, and emergency services response. As such, Wi-Fi locationing represents a significant advancement in leveraging existing wireless network infrastructures to provide added value beyond simple connectivity.

Wi-Fi positioning techniques

Wi-Fi, primarily known for connecting devices and facilitating internet access, also plays a crucial role in location-based services through Wi-Fi positioning techniques. These methods provide valuable geolocation data by analysing the characteristics

DIY wearable location tracker project

Published by the EFY Team, the DIY device is centred around wearable location tracking technology. The device utilises SSID signals alongside magnetic compass data, implemented using the IndusBoard—a versatile IoT device equipped with Wi-Fi capabilities and a magnetic compass sensor. By leveraging Wi-Fi signals and compass data, it achieves real-time location tracking even in challenging environments. For more details, visit: <https://www.electronicsforu.com/electronics-projects/location-tracker-using-ssid-indusboard>



IndusBoard-based location tracker prototype

of Wi-Fi signals, with each technique offering unique benefits and facing distinct challenges.

RSSI positioning. The most common technique, received signal strength indicator (RSSI), uses signal strength measurements for location estimation, primarily through methods like device multilateration or fingerprinting. While RSSI methods are straightforward and cost-effective to implement, their accuracy can be significantly affected by the environment. Objects and people moving within the signal range can cause fluctuations and errors in RSSI readings, leading to less reliable location data.

RSSI multilateration and fingerprinting. RSSI multilateration involves multiple Wi-Fi access points or sensors detecting the signal strength from transmitting devices and using this data to estimate their location. This method is easily activated and inexpensive but is vulnerable to signal-related issues, such as attenuation and interference, which can degrade accuracy.

Fingerprinting, another RSSI-based approach, involves creating a database of the signal strengths at various locations to form a 'fingerprint.' Although this method allows for the location estimation of a device by comparing real-time data to these fingerprints, it requires

extensive initial calibration and ongoing updates to remain accurate, reflecting any environmental changes.

Time of flight and Wi-Fi round-trip time. Time of flight (ToF) represents a more advanced, accuracy-focused technique that measures the time it takes for signals to travel between devices. ToF requires a dense network of accurately synchronised sensors to function effectively. The data from these measurements are analysed to precisely locate devices, offering a higher degree of accuracy than RSSI-based methods. Additionally, Wi-Fi round-trip time (RTT), specified under the IEEE 802.11-2016 standard, calculates the distance between devices based on the time it takes for a signal to make a round trip, providing a foundation for either ranging or multilateration-based positioning.

Angle of arrival. Angle of arrival (AoA) is another sophisticated technique that uses the phase shift detected by a multi-antenna array on a fixed sensor to determine the direction of the incoming signal from a mobile device. AoA can offer more accurate positioning with fewer sensors compared to multilateration, as it requires only two reference points to establish a location, with additional points enhancing precision and reliability.